



北京大学
PEKING UNIVERSITY

Introduction of Enhanced Sampling In Molecular Dynamics

Presenter: Chlo-Hei Lo

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Basic Concept of Enhanced Sampling

Limitation of MD:

- **Imbalance in accuracy and efficiency** of interaction potential.
- The trajectory is **high-dimensional and noisy**
 - Thermodynamics information is hidden.
- Timestep must be **short** while timescales must be long.

Some hard problems for MD:

- Chemical reaction path
- Phase Transition
- Ligand Binding and Unbinding

=> We need method to Explore the **phase space** faster, find the **free energy surface** and figure out the **transition path**.

Basic Concept of Enhanced Sampling

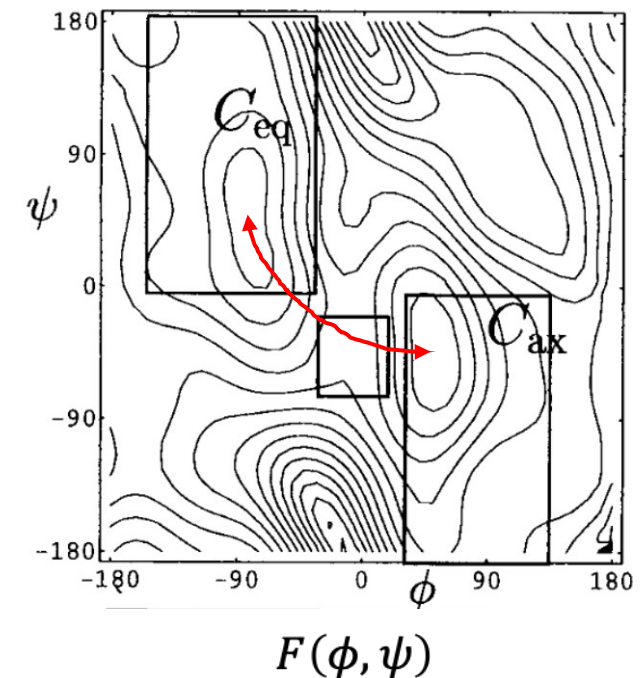
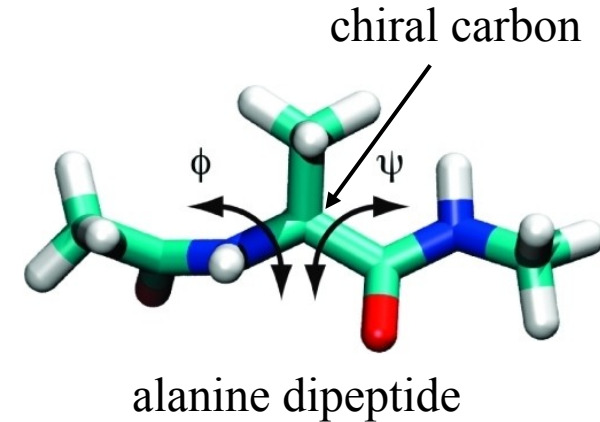
Collective Variables (CVs) : Functions that map from high-dimensional vectors $\mathbf{q} = \{\mathbf{r}_1, \mathbf{r}_2, \dots, \mathbf{r}_{3N}\}$ to low dimensional vector $s_1, s_2, \dots, s_n$, where $n \ll N$.

$$f_i(\mathbf{q}) = s_i$$

CVs or the movement in CVs space R^n carry the thermodynamic properties of the system.

E.g. order parameters, bond length, angles...

The Free Energy Surface with respect to CVs shows excited states and how the transition happens.



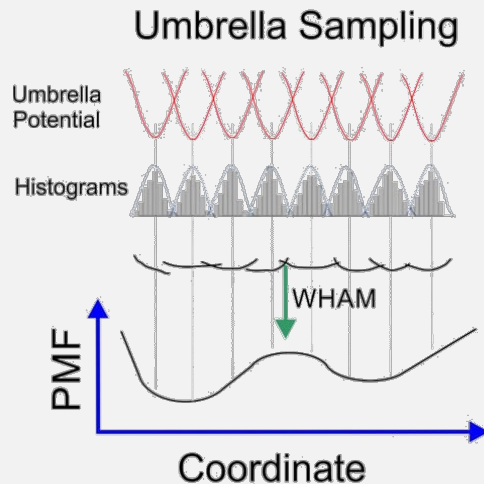
Basic Concept of Enhanced Sampling

The goal of enhanced Sampling is to accelerate the resolution of such Hamiltonian:

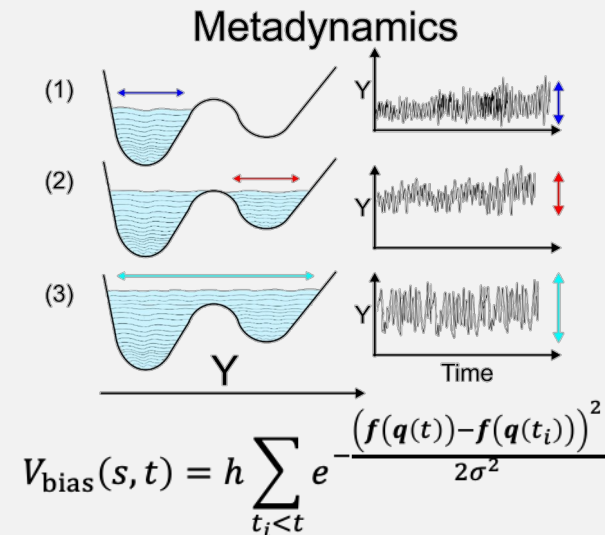
$$H = \sum_i^N \frac{p_i^2}{2m_i} + V(\mathbf{q})$$

Bias-Potential based Methods: (Metadynamics, Umbrella Sampling)

- Modify the interaction by adding $V_{\text{bias}}(f_1(\mathbf{q}), f_2(\mathbf{q}), \dots, f_n(\mathbf{q}))$.



1. Get modified Probability Distribution $P'(s_1, \dots, s_n)$
2. Reweighting $P' \rightarrow P$
3. Estimate the free energy via $F = -kT \ln P$



PART 02

Adiabatic Free Energy Dynamics

Adiabatic Dynamic Theory

Born-Oppenheimer approximation

- Separating the motion of electron and nucleus because $m_e \ll M$.

Adiabatic Dynamic Theory

- The Collective variables could be some “particles” at T_s .
- Different temperature T_s if it does not exchange energy with other particles.
- If $m_s \gg m_i$, the FES could be calculated as:

$$F = -kT_s \ln P_{\text{adb}}(s_1, s_2, \dots, s_n)$$

Adiabatic Dynamic Theory

Probability Distribution of CVs at $s_1, s_2, \dots, s_n$.

$$P(s_1, s_2, \dots, s_n) = \frac{\int \int e^{-\beta H(\mathbf{q}, \mathbf{p})} \prod_{i=1}^n \delta(f_i(\mathbf{q}) - s_i) d\mathbf{q} d\mathbf{p}}{\int \int e^{-\beta H(\mathbf{q}, \mathbf{p})} d\mathbf{q} d\mathbf{p}}.$$

Coordinate Transformation from $\mathbf{q} = \{r_1, \dots, r_{3N}\}$ to $\mathbf{q}' = \{s_1, \dots, s_n, r'_{n+1}, \dots, r'_{3N}\}$.

We can directly sample the distribution P_{adb} from dynamics, and the Hamiltonian becomes:

$$V'(\mathbf{q}') = V(\mathbf{q}(\mathbf{q}')) - kT \ln J(\mathbf{q}')$$

$$H'(\mathbf{q}, \mathbf{p}) = \sum_i^{3N} \frac{p_i^2}{2m'_i} + V'(\mathbf{q}')$$

Assuming $m'_1, \dots, m'_n \gg m'_{n+1}, \dots, m'_{3N}$ and $T_s \neq T$.

AFED Simulation

Select high T_s s.t.

$$kT_s \sim E_{\text{Barrier}}$$

Set $m'_1, \dots, m'_n$ s.t.

$$m'_1, \dots, m'_n \gg m'_{i>n}$$



Transform

$$\mathbf{q} \rightarrow \mathbf{q}'$$



Run two thermostats

$$T_s \text{ for } q'_1, \dots, q'_n$$

$$T \text{ for } q'_{n+1}, \dots, q'_{3N}$$



Output FES

Adiabatic Dynamic Theory

Expand the delta term in gaussian function,

$$\delta(f_i(q) - s_i) = \lim_{\kappa \rightarrow \infty} \sqrt{\frac{\beta\kappa}{2\pi}} \times e^{-\frac{\beta\kappa}{2}(f_i(q) - s_i)^2}$$

The new Hamiltonian becomes,

$$H'(\mathbf{q}, \mathbf{p}, \mathbf{q}_s, \mathbf{p}_s) = \underbrace{\sum_{i=1}^{3N} \frac{p_i^2}{2m_i} + V(\mathbf{q})}_H + \underbrace{\sum_{i=1}^n \frac{p_{s_i}^2}{2m'_i} + \frac{\kappa}{2} \sum_{i=1}^n (f_i(\mathbf{q}) - s_i)^2}_{\text{Extra } n \text{ particles}}$$

Calculated from f_i New "degree of freedom"

The free energy surface is,

$$F = kT_s \lim_{\kappa \rightarrow \infty} \ln P_{\text{adb}}^{(\kappa)}(s_1, s_2, \dots, s_n)$$

d-AFED/TAMD

Simulation

Select high T_s s.t.

$$kT_s \sim E_{\text{Barrier}}$$

Set $\kappa, m'_1, \dots, m'_n$ s.t.

$$m'_1, \dots, m'_n \gg m_{i>n}$$

$$\kappa \gg 1$$

Transform

$$q \rightarrow q' \text{ and } H \rightarrow H'$$

Run two thermostats

$$T_s \text{ for } s_1, \dots, s_n$$

$$T \text{ for } q_1, \dots, q_{3N}$$

Run two thermostats

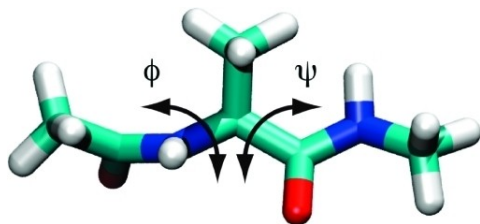
$$T_s \text{ for } q'_1, \dots, q'_n$$

$$T \text{ for } q'_{n+1}, \dots, q'_{3N}$$

Output FES



Simple Example of Alanine Dipeptide



Radius of gyration

$$R_G = \sqrt{\frac{1}{N} \sum_{i=1}^N (\mathbf{r}_i - \frac{1}{N} \sum_{j=1}^N \mathbf{r}_j)^2}$$

Number of H-bonds

$$N_H = \sum_{i \in \mathbf{O}} \sum_{j \in \mathbf{H}} \frac{1 - (\frac{r_i - r_j}{d_0})^6}{1 - (\frac{r_i - r_j}{d_0})^{12}}$$

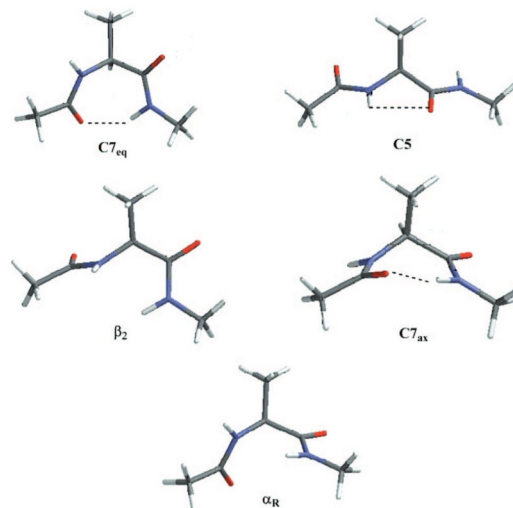
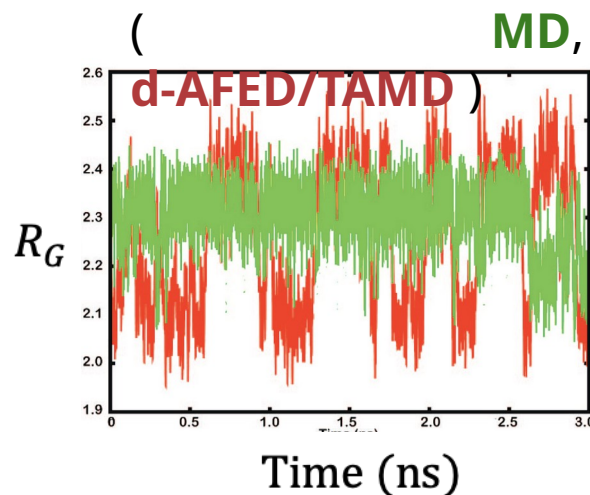
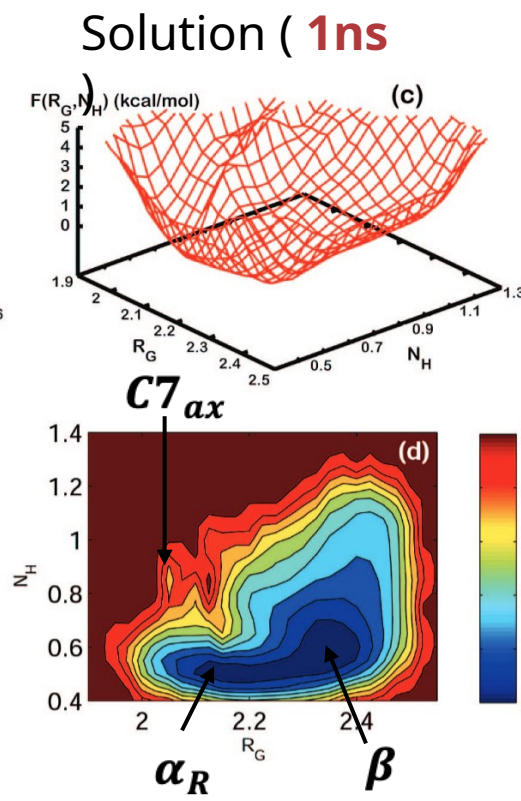
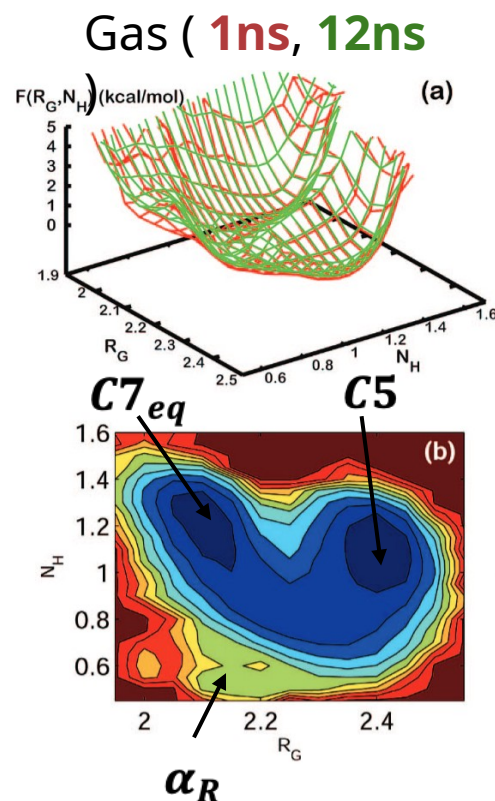


Figure 6. C5, C7_{eq}, C7_{ax}, β₂, and α_R conformations of alanine dipeptide.



Gas phase		
	Experiment	Simulation
ΔF _{C5-C7eq}	0.73	1.0
E _{C5-barrier}	0.93	0.9
Solution phase		
ΔF _{C5-C7eq}	0.73	1.0
E _{C5-barrier}	0.93	0.9



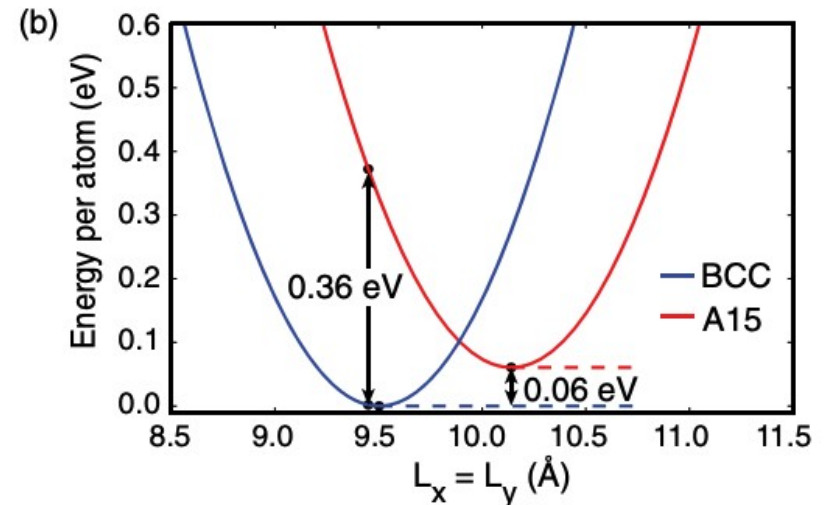
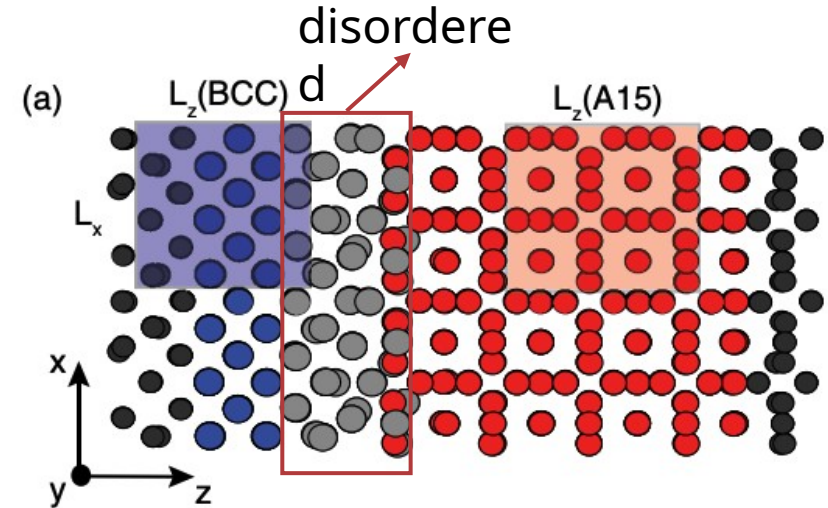
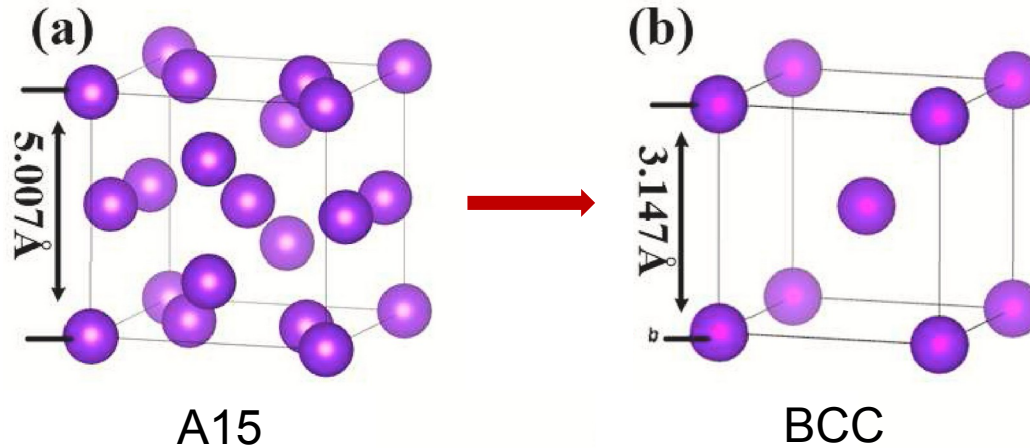
PART 03

Application of Enhanced Sampling

Neural-Network-Based Path CVs of Phase Transformations



A15 Phase (A_3B)
spintronic devices,
high performance material
type II superconductivity



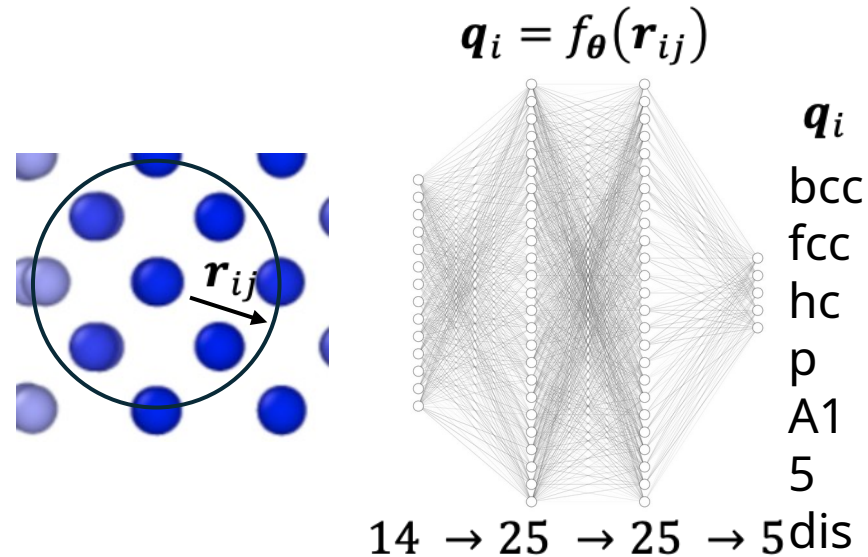
Neural-Network-Based Path CVs of Phase Transformations

Maps local environment $\mathbf{r}_{ij} \rightarrow \mathbf{R}^{14}$ using three Steinhardt parameters q_6, q_7, q_8 and 11 symmetry functions.

$$G_2^{(i)}(\mathbf{r}_{ij}) = \sum_{j \neq i} e^{-\eta(|\mathbf{r}_{ij}| - R_s)^2} f_c(\mathbf{r}_{ij}),$$

$$G_3^{(i)}(\mathbf{r}_{ij}) = \sum_{j \neq i} \cos(\kappa|\mathbf{r}_{ij}|) f_c(\mathbf{r}_{ij}).$$

Training data includes bcc, fcc, hcp, A15
MD data from 300K to 6000K.



Global
attribute:

$$Q = \frac{1}{N} \sum_i q_i$$

Neural-Network-Based Path CVs of Phase Transformations

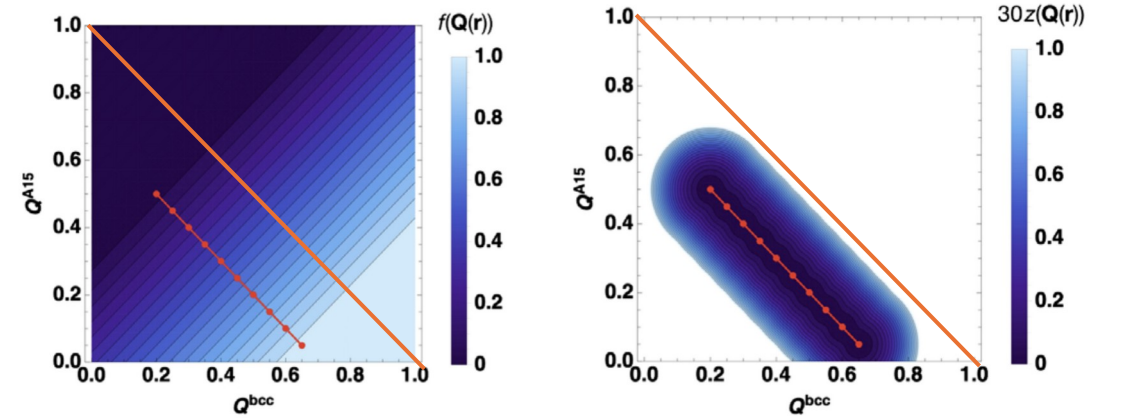
Constructing the path of length N_{path} ,

$$\mathbf{Q}_k \in \{\mathbf{Q}_{\text{bcc}}, \mathbf{Q}_{\text{A15}}\}$$

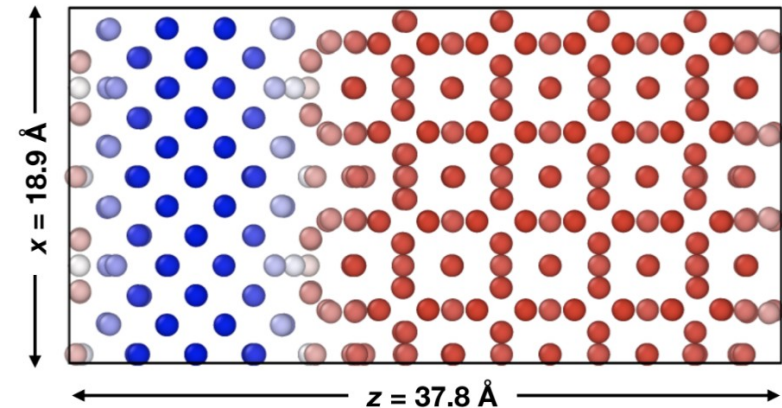
A 1D path CV f_{path} ,

$$f_{\text{path}}(\mathbf{Q}) = \frac{1}{N_{\text{path}} - 1} \frac{\sum_{k=1}^{N_{\text{path}}} (k - 1) e^{-\lambda |\mathbf{Q} - \mathbf{Q}_k|^2}}{\sum_{k=1}^{N_{\text{path}}} e^{-\lambda |\mathbf{Q} - \mathbf{Q}_k|^2}}$$

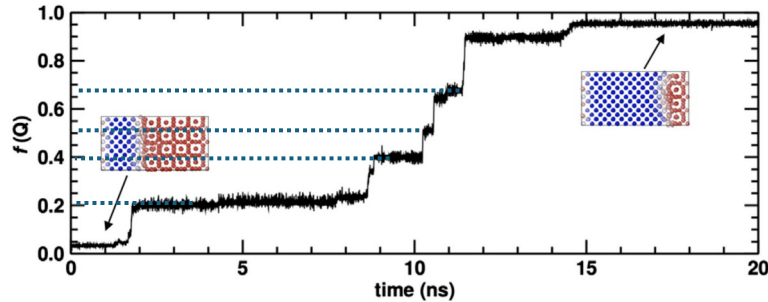
For NPT simulation, a constrain potential is added based on the distance to the path.



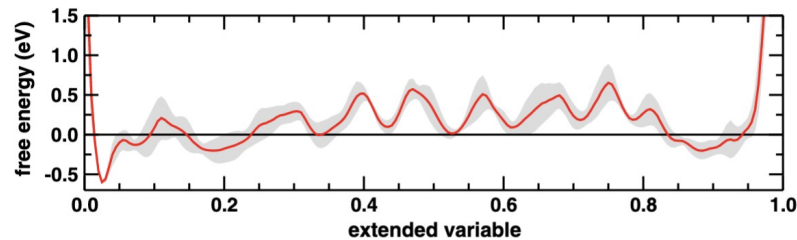
$$z(\mathbf{Q}) = -\frac{1}{\lambda} \ln \sum_{k=1}^{N_{\text{path}}} e^{-\lambda |\mathbf{Q} - \mathbf{Q}_k|^2}$$



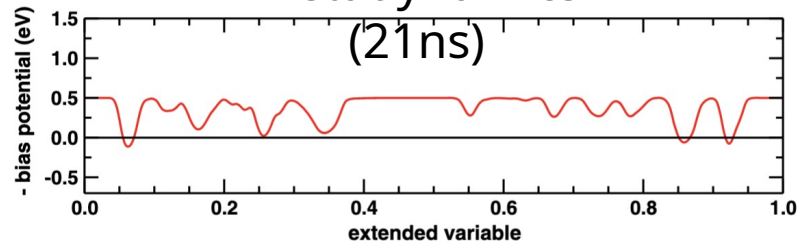
Neural-Network-Based Path CVs of Phase Transformations



d-AFED / TAMD



Metadynamics



Using d-AFED / TAMD and MetaD, the system in NVT ensemble converts from A15 to bcc using only < 20 ns, while no-bias simulation needs ~ 10 ms.

Here, since only forward process is observed, (mismatch of crystal)
The mean force for s is $F'(s) = -\langle \kappa(f(\mathbf{Q}) - s) \rangle_s^{\text{adb}}$

How can the author integrate the force without specify starting point?

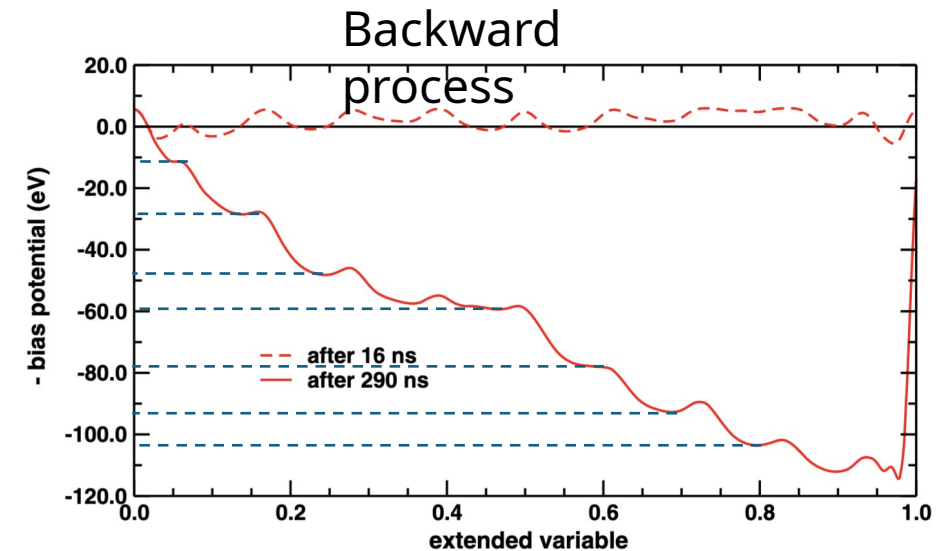
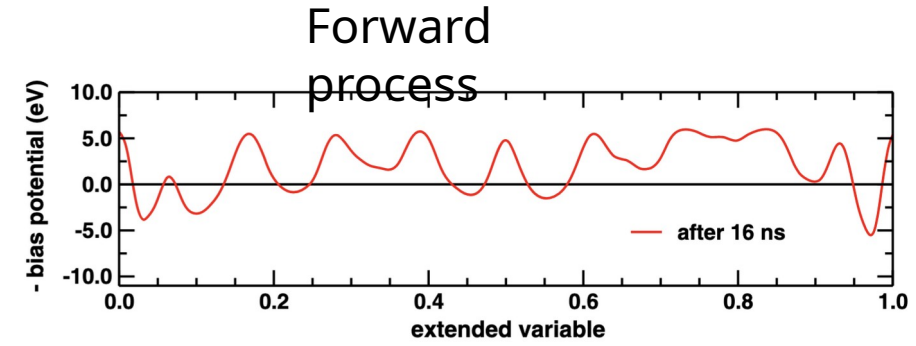
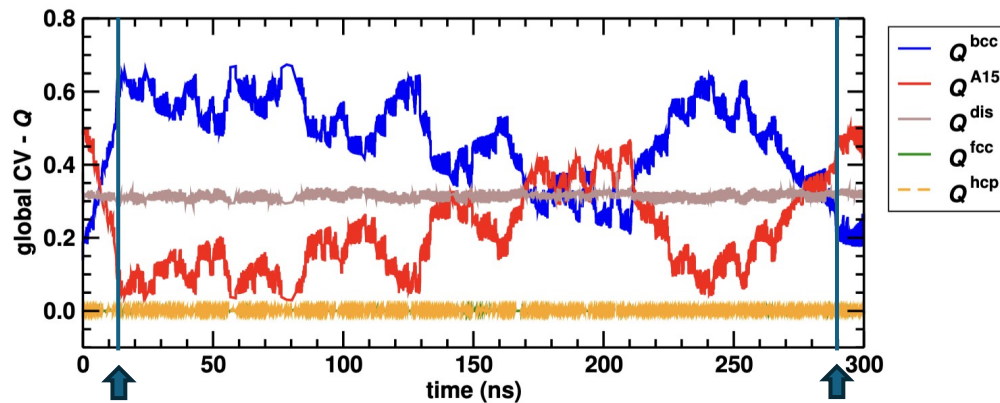
For both MetaD and d-AFED / TAMD, giving the free energy around 0.5 eV, aligns with previous work using Adaptive Kinetic Monte Carlo about 0.47 ± 0.07 eV.

Neural-Network-Based Path CVs of Phase Transformations

In NPT ensemble, the simulation only runs in MetaD.

The forward barrier is much larger because no crystal strain provides energy here.

Reversed process looks like a step function because it's consuming energy.



Conclusion

- Enhanced Sampling not only accelerates MD but also provides methods to estimate FES.
- Slow process could be detangled from the system by creating a heavy particle representing CVs.
- Combining NNs and Enhanced Sampling, We could model realistic interface mismatches and transformation mechanisms more accurately.